MSG No. 53.333 P_Amna [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `2'22'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[1,3,10,12]`
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[2,4,9,11]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[5,7,14,16]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[6,8,13,15]`

Table 2: Wyckoff site: `4b`, site symmetry: `2'22'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[1,3,10,12]`
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[2,4,9,11]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[5,7,14,16]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[6,8,13,15]`

Table 3: Wyckoff site: `4c`, site symmetry: `2'/m..`

No.	position	mapping
1	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	`[1,6,10,13]`
2	$\left[0, \frac{1}{4}, \frac{1}{4}\right]$	`[2,5,9,14]`
3	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	`[3,8,12,15]`
4	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	`[4,7,11,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `2'/m..`

No.	position	mapping
1	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[1,6,10,13]`
2	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[2,5,9,14]`
3	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[3,8,12,15]`
4	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[4,7,11,16]`

Table 5: Wyckoff site: `4e`, site symmetry: `2/m..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,5,6]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,4,7,8]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[9,10,13,14]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[11,12,15,16]`

Table 6: Wyckoff site: `4f`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,5,6]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[3,4,7,8]`
3	$\left[0,\ \frac{1}{2},\ 0\right]$	`[9,10,13,14]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[11,12,15,16]`

Table 7: Wyckoff site: `8g`, site symmetry: `.2.`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ \frac{3}{4}\right]$	`[1,3]`
2	$\left[\frac{1}{4},\ -y,\ \frac{1}{4}\right]$	`[2,4]`
3	$\left[\frac{3}{4},\ -y,\ \frac{1}{4}\right]$	`[5,7]`
4	$\left[\frac{3}{4},\ y,\ \frac{3}{4}\right]$	`[6,8]`
5	$\left[\frac{1}{4},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[9,11]`
6	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[10,12]`
7	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[13,15]`
8	$\left[\frac{3}{4},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[14,16]`

Table 8: Wyckoff site: `8h`, site symmetry: `..2'`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[1,12]`
2	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	`[2,11]`
3	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[3,10]`
4	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[4,9]`
5	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[5,16]`
6	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	`[6,15]`
7	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[7,14]`
8	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[8,13]`

Table 9: Wyckoff site: `8i`, site symmetry: `2'..`

No.	position	mapping
1	$\left[x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[1,10]`
2	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[2,9]`
3	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[3,12]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[4,11]`
5	$\left[-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[5,14]`
6	$\left[-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[6,13]`
7	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[7,16]`
8	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[8,15]`

Table 10: Wyckoff site: `8j`, site symmetry: `2'..`

No.	position	mapping
1	$\left[x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[1,10]`
2	$\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[2,9]`
3	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[3,12]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[4,11]`
5	$\left[-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[5,14]`
6	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[6,13]`
7	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[7,16]`
8	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[8,15]`

Table 11: Wyckoff site: `8k`, site symmetry: `2..`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,2]`
2	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	`[3,4]`
3	$[-x,\ 0,\ 0]$	`[5,6]`
4	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[7,8]`
5	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,10]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	`[11,12]`
7	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[13,14]`
8	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[15,16]`

Table 12: Wyckoff site: `8l`, site symmetry: `m..`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,6]`
2	$[0,\ -y,\ -z]$	`[2,5]`
3	$\left[\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	`[3,8]`
4	$\left[\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	`[4,7]`
5	$\left[0,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,14]`
6	$\left[0,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[10,13]`
7	$\left[\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	`[11,16]`
8	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	`[12,15]`

Table 13: Wyckoff site: `16m`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[x,\ -y,\ -z]$	`[2]`
3	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	`[3]`
4	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	`[4]`
5	$[-x,\ -y,\ -z]$	`[5]`
6	$[-x,\ y,\ z]$	`[6]`
7	$\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	`[7]`

continued ...

Table 13

No.	position	mapping
8	$\left[x+\frac{1}{2},\, y,\, \frac{1}{2}-z\right]$	[8]
9	$\left[x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[9]
10	$\left[x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[10]
11	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	[11]
12	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[12]
13	$\left[-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[13]
14	$\left[-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[14]
15	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	[15]
16	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	[16]